PREP 3, LESSONS 5–6

Wires, series and parallel

Resistance of a wire (required practical); rules for series and parallel circuits

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 State the standard units of potential difference and resistance. DN 96 [1]

Potential difference: volt (V); resistance: ohm (Ω)

2 Describe how current depends on potential difference for an ohmic conductor. DN 98 [1]

Current is directly proportional to potential difference.

3 Describe the current at different points in a single closed loop. DN 92 [1]

It is the same at every point.

Key ideas LESSONS 5–6

Show your working

4 A student measures the resistance of a wire at different lengths.

Length / cm	20	40	60	80
Resistance / Ω	0.8	1.6	2.4	3.2

a Describe the relationship between resistance and length. [1]

Resistance is directly proportional to length (doubling the length doubles R) ✓

b Name one control variable. [1]

Thickness (diameter) of the wire, material, or temperature ✓

c Explain why the potential difference is kept small. [1]

So the current is small and the wire does not heat up (its resistance would change) ✓

5 Two lamps are connected in series to a 12 V battery.

a Calculate the reading on V_2. [1]

12 − 7.5 = 4.5 V ✓

b Two resistors, 4 Ω and 8 Ω, are in series. Total resistance: 12 Ω [1]

c Describe the potential difference across components in parallel. [1]

It is the same across each component ✓

d In a parallel circuit, the total current is 0.90 A. One branch has 0.40 A. The other branch has 0.50 A [1]

